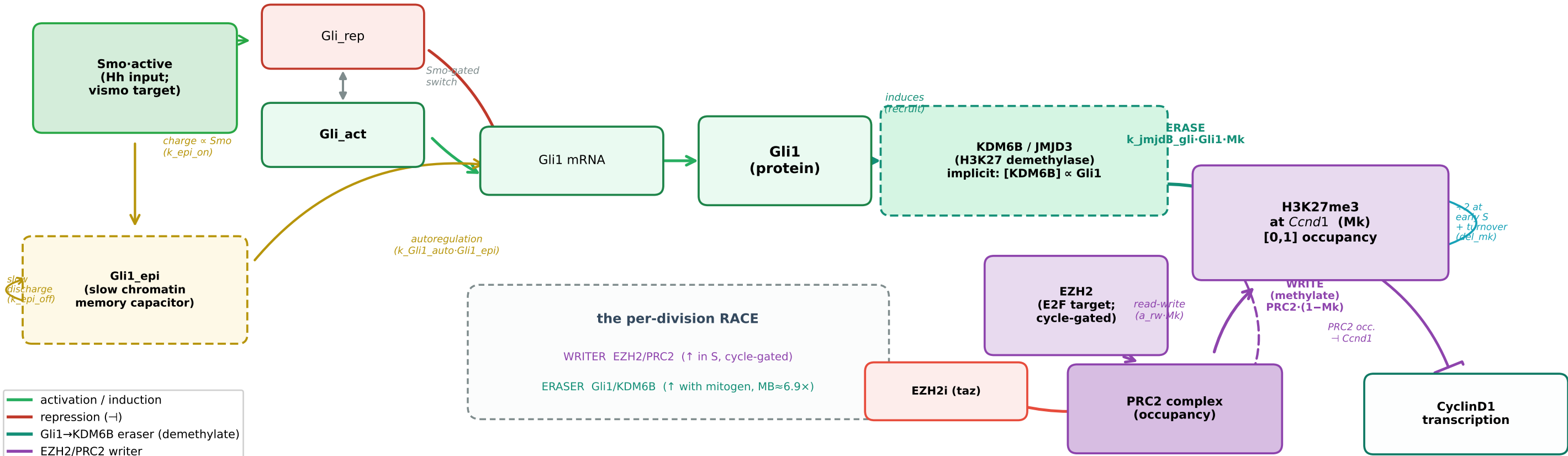


Gli1 → Kdm6b/Jmjd3 eraser of H3K27me3 at *Ccnd1* — detailed wiring + equations (v44.1)

mitogen-gated ERASER (Gli1 → KDM6B/JMJD3 demethylase) vs cycle-gated WRITER (EZH2/PRC2) — the per-division race that sets the mark



① Gli1 axis (mitogen input, with slow autoregulatory memory)

$\text{Gli_rep} \rightleftharpoons \text{Gli_act}$ (Smo-gated activator/repressor switch)

$$\frac{d[\text{Gli1_mRNA}]}{dt} = V_{\text{max}} \cdot \text{Gli_act}^2 / (K_{\text{GA}}^2 + \text{Gli_act}^2) \cdot (1 - \text{Gli_rep}^2 / (K_{\text{GR}}^2 + \text{Gli_rep}^2)) \cdot \gamma_{\text{Ptch}}$$
$$+ k_{\text{Gli1_auto}} \cdot \text{Gli1_epi} - k_{\text{mdeg}} \cdot \text{Gli1_mRNA}$$
$$\frac{d[\text{Gli1_epi}]}{dt} = k_{\text{epi_on}} \cdot (1 - \text{Ptch}) \cdot \text{Smo}^4 / (K_{\text{auto}}^4 + \text{Smo}^4) \cdot (1 - \text{Gli1_epi}) - k_{\text{epi_off}} \cdot \text{Gli1_epi}$$
$$\frac{d[\text{Gli1}]}{dt} = k_{\text{Gli1_tl}} \cdot \text{Gli1_mRNA} - k_{\text{Gli1_deg}} \cdot \text{Gli1}$$

② Eraser (KDM6B/JMJD3, implicit)

KDM6B is NOT a state variable.
Quasi-static: $[\text{KDM6B}] \propto \text{Gli1}$,
folded into $k_{\text{jmd3_gli}}$.

erase flux =
 $k_{\text{jmd3_gli}} \cdot \text{Gli1} \cdot \text{Mk}$

③ H3K27me3 (Mk) balance at the *Ccnd1* domain

$$\frac{d[\text{Mk}]}{dt} = \underbrace{\text{PRC2} \cdot (1 - \text{Mk})}_{\text{(EZH2/PRC2, cycle)}} - \underbrace{\text{del_mk} \cdot \text{Mk}}_{\text{(Gli1/KDM6B, mitogen)}} - \underbrace{k_{\text{jmd3_gli}} \cdot \text{Gli1} \cdot \text{Mk}}_{\text{ERASE}}$$

discrete: at $\text{Dna} > 0.05 \rightarrow \text{Mk} \leftarrow \frac{1}{2} \cdot \text{Mk}$ (replicative dilution, once per S)

$$\text{PRC2} = \text{EZH2} \cdot (1 - \text{EZH2i}) \cdot (a_0 + a_{\text{rw}} \cdot \text{Mk}) \cdot (1 - g \cdot \text{Cd_mRNA}^p / (K_{\text{tx}}^p + \text{Cd_mRNA}^p))$$

④ Baked parameters (v44.1) + net effect

$k_{\text{jmd3_gli}}$ 0.0478 del_mk 0.0015
 a_0 3.78e-4 a_{rw} 4.30e-3 (a_{rw}/a_0 11.4)
 g_{prc2} 0.0424 K_{prc2} 3.5e-3 n 3.39
 f_0_{prc2} 0.0505 K_{tx} 5.12 p_{tx} 3.27
Gli: V_{max} 1.27 auto 0.012 on/off 0.02/8e-4

**MB Gli1 ≈ 6.9× GNP → erase ≈ 6.9× stronger →
ChIP *Ccnd1* mark MB/GNP ≈ 0.55 (MB < GNP)**

Net over divisions: *Mk* accumulates when $\text{WRITE} > (\text{turnover} + \text{ERASE} + \frac{1}{2}\text{-dilution})$ each cycle; MB's high Gli1 tilts the race toward erasure despite higher EZH2.